

Name:Supriya.B Register Number: 250701760 Experiment Name:K-Means Clustering

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df=pd.read_csv('Mall_Customers.csv')

df.info()

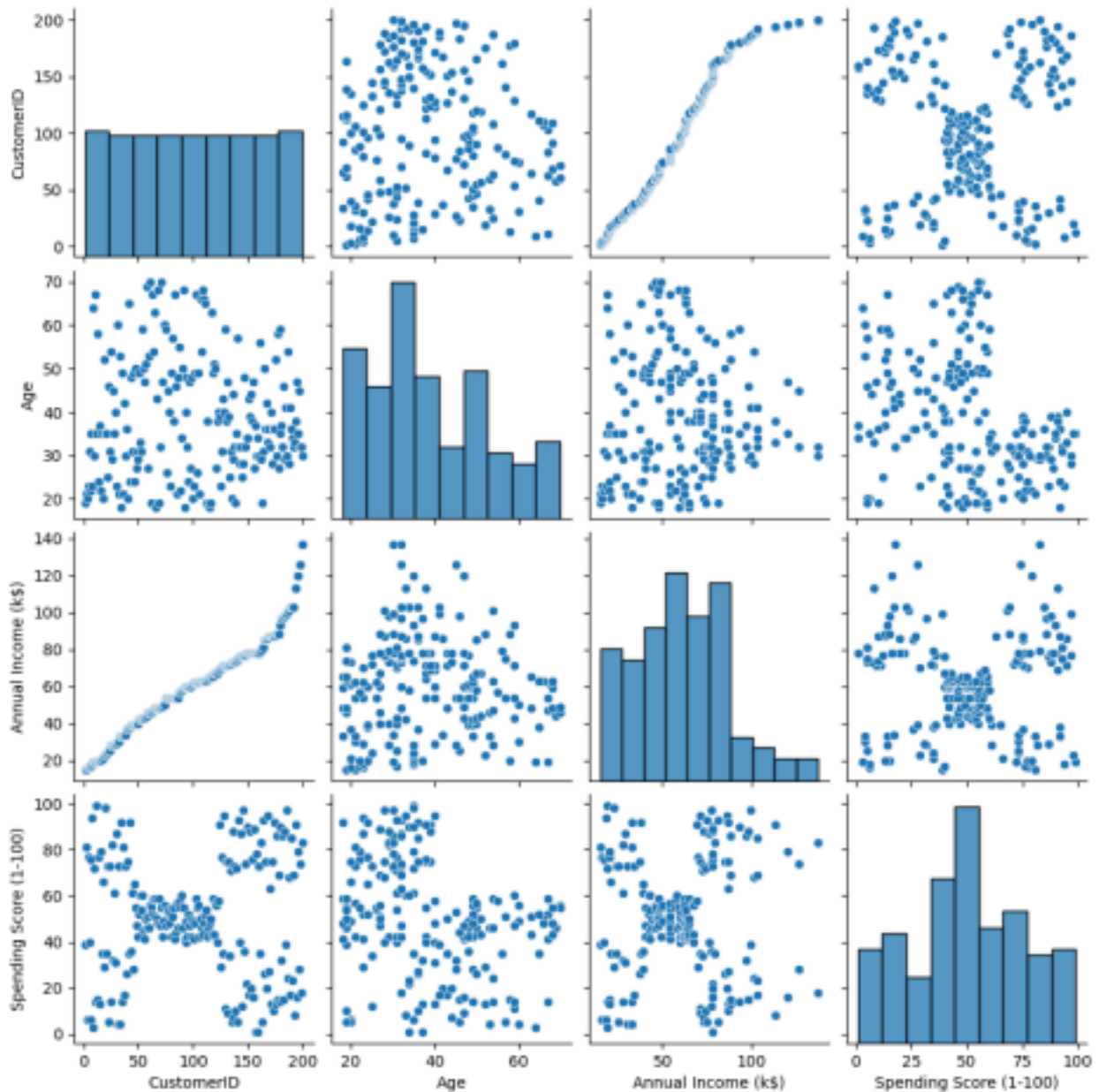
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
# Column Non-Null Count Dtype
-----
0 CustomerID 200 non-null int64
1 Gender 200 non-null object
2 Age 200 non-null int64
3 Annual Income (k$) 200 non-null int64
4 Spending Score (1-100) 200 non-null int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

df.head()

CustomerID Gender Age Annual Income (k$) Spending Score (1-100) 0 1 Male 19 15 39 1 2
Male 21 15 81 2 3 Female 20 16 6 3 4 Female 23 16 77 4 5 Female 31 17 40

sns.pairplot(df)

<seaborn.axisgrid.PairGrid at 0x1f67ce53230>
```



```
features=df.iloc[:,[3,4]].values
```

```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

C:\Users\swath\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there
are less chunks than available threads. You can avoid it by setting the environment variable

```
OMP_NUM_THREADS=1.
```

```
warnings.warn(
```

```
KMeans(n_clusters=5)
```

```
Final=df.iloc[:,[3,4]]
```

```
Final["label"]=model.predict(features)
```

```
Final.head()
```

```
C:\Users\swath\AppData\Local\Temp\ipykernel_5828\470183701.py:2:
```

```
SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame. Try using
```

```
.loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation:
```

```
https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#
```

```
returning-a-view-versus-a-copy
```

```
Final["label"]=model.predict(features)
```

```
Annual Income (k$) Spending Score (1-100) label
```

```
0 15 39 4
```

```
1 15 81 3
```

```
2 16 6 4
```

```
3 16 77 3
```

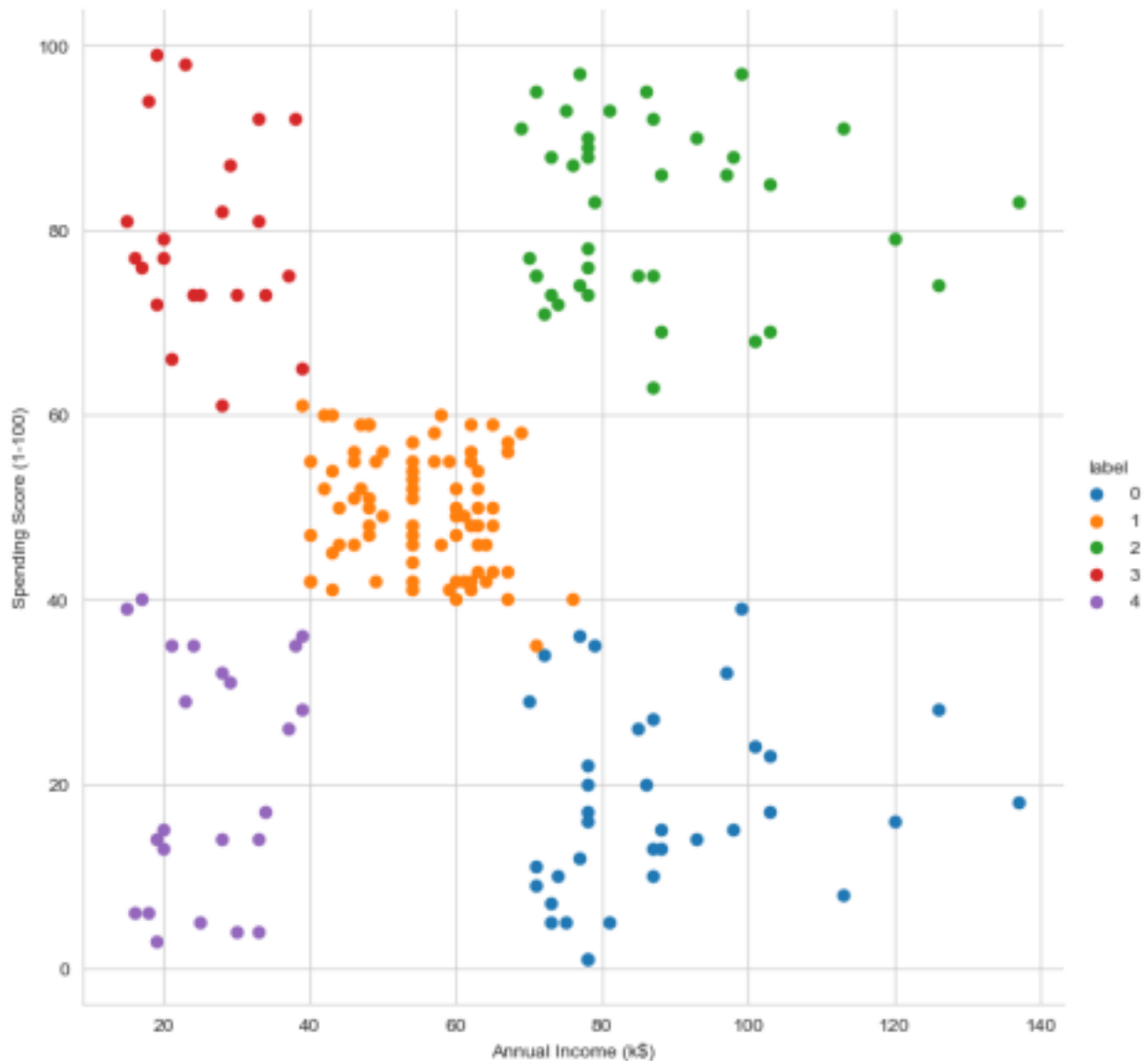
```
4 17 40 4
```

```
sns.set_style("whitegrid")
```

```
sns.FacetGrid(Final,hue="label",height=8) \
```

```
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \ .add_legend();
```

```
plt.show()
```



```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Cell In[11], line 5

```
model=KMeans(n_clusters=i)
```

^

IndentationError: expected an indented block after 'for' statement on line 4

